Requirement Analysis Document: Online Shopping Platform

Week 1 – Understanding the Business Problem

1. Introduction

1.1 Purpose

This document analyses how large online shopping platforms such as Amazon and Flipkart operate, and defines the functional and data requirements for a proposed e-commerce system. It will serve as the basis for database design (ER modelling, normalization and schema creation) in later weeks.

1.2 Scope

The proposed system lets customers register, browse and search products, manage a cart, place orders, pay online, and track shipments. It also lets sellers list and manage products, and lets administrators oversee the platform. Out of scope for this phase: recommendation engines, advertising, and mobile-app UI design.

1.3 Definitions

Term	Meaning
Customer	A registered user who buys products
Seller	A vendor who lists and sells products on the platform
SKU	Stock Keeping Unit, a unique identifier for a sellable product variant
Order	A confirmed purchase request containing one or more items
Shipment	The physical delivery of all or part of an order

2. How an Online Shopping Platform Works

An e-commerce platform is a marketplace connecting customers, sellers and logistics partners. The typical flow is:

1. A visitor creates an account and logs in.
2. The customer searches or browses categories, filters results, and views product details, prices, ratings and reviews.
3. Chosen items go into a cart (or wishlist), where quantities and prices are reviewed.
4. At checkout the customer selects a delivery address and a payment method, then places the order.
5. The payment gateway authorises the payment; the system confirms the order and reduces stock.
6. The seller packs the items, a courier picks them up, and the shipment moves through dispatch, in-transit, out-for-delivery and delivered stages.
7. After delivery the customer can rate and review the product, or request a return, replacement or refund.

3. Stakeholders and Users

User	Role in the system
Customer	Registers, browses, orders, pays, tracks, reviews, returns
Seller	Manages product listings, prices, stock, and fulfils orders
Administrator	Manages users, categories, disputes, reports and platform policies
Delivery partner	Updates shipment status and delivery confirmation
Payment gateway (external)	Processes and confirms payments and refunds

4. Major Activities in the System

4.1 Customer Registration and Account Management

New users sign up with name, email, phone number and password. Customers can log in, update their profile, and manage multiple delivery addresses. Email or OTP verification confirms the identity.

4.2 Product Browsing and Search

Products are organised by category and subcategory. Customers can search by keyword, filter by price, brand, rating and availability, sort results, and open a product page showing description, images, price, stock status, seller and reviews.

4.3 Cart and Wishlist

Customers add, remove or change the quantity of items in a cart, and save items to a wishlist for later. The cart shows running totals, taxes, discounts and delivery charges.

4.4 Order Placement

At checkout the customer confirms items, delivery address and payment method. The system validates stock, calculates the final amount (including taxes, coupons and shipping), creates an order with a unique order ID, and records each item with the price at the time of purchase.

4.5 Payment Processing

Supported methods include cards, UPI, net banking, wallets and cash on delivery. Each payment attempt is recorded with status (pending, successful, failed, refunded), transaction reference, amount and timestamp. Failed payments do not confirm the order; refunds are issued for cancellations and approved returns.

4.6 Shipment and Delivery Tracking

Once an order is confirmed, one or more shipments are created. Each shipment has a courier, tracking number, and status history with timestamps. Customers can view live status and the expected delivery date.

4.7 Cancellation, Returns and Refunds

Customers can cancel orders before dispatch and request returns or replacements within the return window. Approved returns trigger a pickup and a refund to the original payment method.

4.8 Reviews and Ratings

Customers who purchased a product may rate it and write a review. Average ratings are shown on product pages.

4.9 Seller and Inventory Management

Sellers add products, set prices and discounts, update stock, view their orders, and mark orders as packed or shipped.

4.10 Administration and Reporting

Administrators manage categories, approve sellers, handle complaints, and generate reports such as daily sales, top-selling products, low-stock items and revenue by category.

5. Functional Requirements

Customer Registration and Login

- **FR1.** The system shall allow a new customer to register with a unique email and phone number.
- **FR2.** The system shall authenticate customers using email/phone and password.
- **FR3.** The system shall allow customers to add, edit and delete multiple delivery addresses.
- **FR4.** The system shall allow customers to update their profile and reset their password.

Product Catalogue

- **FR5.** The system shall store products with name, description, brand, category, price, stock quantity and seller.

- **FR6.** The system shall support searching products by keyword and filtering by category, price range, brand and rating.
- **FR7.** The system shall display product availability and show "out of stock" when quantity is zero.

Cart and Wishlist

- **FR8.** The system shall allow customers to add products to a cart, change quantities, and remove items.
- **FR9.** The system shall allow customers to maintain a wishlist.

Orders

- **FR10.** The system shall create an order with a unique ID, date, customer, delivery address and total amount.
- **FR11.** The system shall store each ordered item with quantity and unit price at the time of purchase.
- **FR12.** The system shall reduce stock when an order is confirmed and restore it on cancellation.
- **FR13.** The system shall support applying valid discount coupons to an order.
- **FR14.** The system shall allow customers to view their order history and order details.
- **FR15.** The system shall allow cancellation of an order that has not yet been shipped.

Payments

- **FR16.** The system shall allow payment by multiple methods (card, UPI, net banking, wallet, cash on delivery).
- **FR17.** The system shall record every payment with method, amount, status, transaction ID and timestamp.
- **FR18.** The system shall confirm an order only after successful payment (or selection of cash on delivery).
- **FR19.** The system shall record refunds against the original payment.

Shipment Tracking

- **FR20.** The system shall create a shipment for each confirmed order, with a courier and tracking number.
- **FR21.** The system shall maintain a timestamped status history for each shipment.

- **FR22.** The system shall let customers track the current status and expected delivery date of their order.

Returns, Reviews and Notifications

- **FR23.** The system shall allow return or replacement requests for delivered items within the return window.
- **FR24.** The system shall allow only customers who bought a product to rate and review it.
- **FR25.** The system shall send notifications (email/SMS) for registration, order confirmation, payment status, dispatch and delivery.

Seller and Admin

- **FR26.** The system shall allow sellers to add, update and deactivate their product listings and stock.
- **FR27.** The system shall allow administrators to manage users, sellers, categories and complaints.
- **FR28.** The system shall generate sales, inventory and revenue reports.

6. Business Rules

1. A customer must be registered and logged in to place an order.
2. Email and phone number must be unique per customer.
3. An order contains at least one item, and each item belongs to exactly one seller.
4. An item cannot be ordered if requested quantity exceeds available stock.
5. The price stored in an order item is fixed at purchase time, even if the product price changes later.
6. An order can have one or more payment attempts, but only one successful payment per order.
7. An order can be cancelled only before it is shipped.
8. A review may be written only for a product the customer has received.
9. Refunds go to the original payment method, except for cash on delivery orders.

7. Preliminary Data Requirements

The following entities have been identified from the analysis. They will be refined into an ER diagram in the next phase.

Entity	Key information stored
Customer	ID, name, email, phone, password hash, registration date
Address	ID, customer, street, city, state, pin code, type
Category	ID, name, parent category
Seller	ID, name, contact details, rating
Product	ID, name, description, brand, price, stock, category, seller
Cart / Cart Item	Customer, product, quantity, date added
Wishlist	Customer, product
Order	ID, customer, address, date, status, total amount
Order Item	Order, product, quantity, unit price
Payment	ID, order, method, amount, status, transaction ID, date
Shipment	ID, order, courier, tracking number, status, expected delivery date
Shipment Status History	Shipment, status, location, timestamp
Return / Refund	Order item, reason, status, refund amount
Review	Customer, product, rating, comment, date
Coupon	Code, discount value, validity period, usage limit

Key relationships: a customer places many orders; an order contains many order items; a product appears in many order items; an order has one or more payments and shipments; a seller lists many products; a category contains many products; a customer writes many reviews.

8. Non-Functional Requirements

- **Security:** passwords stored hashed; payment details never stored in plain text; role-based access for customers, sellers and admins.
- **Performance:** search results and product pages should load within a few seconds, even with a large catalogue.
- **Data integrity:** orders, payments and stock updates must be handled as atomic transactions so that no order is confirmed without payment and stock is never over-sold.
- **Availability and scalability:** the system should remain usable during high-traffic periods such as festival sales.
- **Usability:** simple navigation and a clear checkout process.
- **Auditability:** order, payment and shipment histories are retained for reporting and dispute resolution.

9. Assumptions and Constraints

- Payments are processed by a third-party gateway; the system stores only the transaction reference and status.
- Delivery is carried out by external courier partners who supply status updates.
- Prices are in a single currency (INR) and taxes are applied as a percentage on the product price.
- The initial version supports one delivery address per order and one seller per order item.

10. Conclusion

The analysis shows that an online shopping system is built around five core flows: registration, browsing, order placement, payment and shipment tracking, supported by seller, returns and reporting functions. The functional requirements, business rules and preliminary entities above provide the foundation for the next phase, which is to build the ER diagram and convert it into a relational schema.